PCM INSPECTION [SKYACTIV-G 1.5, SKYACTIV-G 2.0]

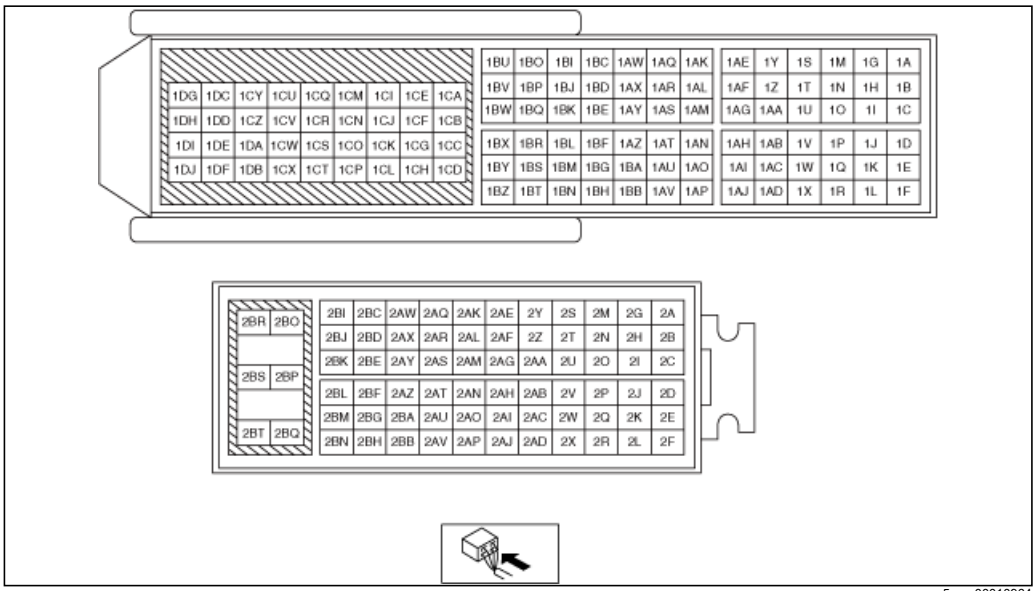
id0140g8802500

Without Using the M-MDS

Note

- Because the PCM uses a waterproof connector, the inspection for the voltage/wave pattern cannot be performed. The following values are for reference.

Terminal voltage table (Reference)



ac5wzw00010964

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1A*1	CAN_2H	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
1B*1	CAN_2L	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
1C	—	—	—		—	—
1D	—	—	—		—	—
1E	Constant voltage (Vref)	CKP sensor	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
1F	CKP	CKP sensor	(See CKP signal.)			• CKP sensor • Related wiring harness
1G	—	—	—		—	—
1H	—	—	—		—	—
1I	—	—	—		—	—
1J	Constant voltage (Vref)	MAP sensor	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
1K	Constant voltage (Vref)	Fuel pressure sensor	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
1L	Constant voltage (Vref)	TP sensor No.1, TP sensor No.2	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
1M	A/F	A/F sensor	Idle (after warm up)		Approx. 3.99	• A/F sensor • Related wiring harness
1N	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness
1O	TP (No.1)	TP sensor No.1	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 1.0	• TP sensor No.1 • Related wiring harness
				Accelerator pedal depressed	Approx. 4.61	
1P	—	—	—		—	—
1Q	IAT (No.2)	IAT sensor No.2	Ignition switched ON (engine off)	IAT 0 °C {32 °F}	Approx. 5.9	• IAT sensor No.2 • Related wiring harness
				IAT 20 °C {68 °F}	Approx. 2.5	
				IAT 40 °C {104 °F}	Approx. 0.323	
				IAT 60 °C {140 °F}	Approx. 0.0893	
1R	GND	TP sensor No.1, TP sensor No.2	Under any condition		Below 1.0	• Related wiring harness
1S	A/F	A/F sensor	Idle (after warm up): 0 mA			• A/F sensor • Related wiring harness
1T	A/F	A/F sensor	Idle (after warm up)		Approx. 4.12	• A/F sensor • Related wiring harness
1U	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1V	TP (No.2)	TP sensor No.2	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 4.31	• TP sensor No.2 • Related wiring harness
				Accelerator pedal depressed	Below 1.0	
1W	Fuel pressure	Fuel pressure sensor	Ignition switched ON (engine off)		Approx. 5.01	• Fuel pressure sensor • Related wiring harness
			Idle (after warm up)		Approx. 1.51	
1X	MAP	MAP sensor	Ignition switched ON (engine off)		Approx. 5.02	• MAP sensor • Related wiring harness
			Idle (after warm up)		Approx. 1.80	
			Racing (Engine speed: 2,000 rpm)		Approx. 1.18	
1Y	Knocking (+)	KS	Ignition switched ON (engine off) (Use digital type voltmeter, because measurement voltage will be detected less than true voltage when using analog type voltmeter)		Approx. 2.44	• KS • Related wiring harness
1Z	Knocking (–)	KS	Ignition switched ON (engine off) (Use digital type voltmeter, because measurement voltage will be detected less than true voltage when using analog type voltmeter)		Below 1.0	• KS • Related wiring harness
1AA	—	—	—		—	—
1AB	GND	ECT sensor	Ignition switched ON (engine off)		Below 1.0	• Related wiring harness
1AC	—	—	—		—	—
1AD	Electric variable valve timing driver (diagnostic)	Electric variable valve timing driver	(See Electric variable valve timing driver (diagnostic) signal.)			• Electric variable valve timing motor/driver • Related wiring harness
1AE	Ion (No.2)	Ion sensor No.2	Idle (after warm up)		Approx. 4.44	• Ion sensor No.2 • Related wiring harness
1AF	Ion (No.4)	Ion sensor No.4	Idle (after warm up)		Approx. 4.44	• Ion sensor No.4 • Related wiring harness
1AG*4	Generator output voltage	Generator	(See Generator output voltage (without i-ELOOP).)			• Generator • Related wiring harness
1AH	ECT	ECT sensor	Ignition switched ON (engine off)	ECT 20 °C {68 °F}	Approx. 3.10	• ECT sensor • Related wiring harness
				ECT 40 °C {104 °F}	Approx. 2.16	
				ECT 60 °C {140 °F}	Approx. 1.40	
				ECT 80 °C {176 °F}	Approx. 0.87	
				ECT 100 °C {212 °F}	Approx. 0.54	
1AI	GND	Intake CMP sensor, exhaust CMP sensor	Under any condition		Below 1.0	• Related wiring harness
1AJ	—	—	—		—	—
1AK	Ion (No.3)	Ion sensor No.3	Idle (after warm up)		Approx. 4.44	• Ion sensor No.3 • Related wiring harness
1AL	Ion (No.1)	Ion sensor No.1	Idle (after warm up)		Approx. 4.44	• Ion sensor No.1 • Related wiring harness
1AM	—	—	—		—	—
1AN	—	—	—		—	—
1AO	—	—	—		—	—
1AP	GND	CKP sensor	Ignition switched ON (engine off)		Below 1.0	• Related wiring harness
1AQ*3	Generator field coil control	Generator	(See Generator field coil control signal (with i-ELOOP).)			• Generator • Related wiring harness
1AR*3	Generator output voltage	Generator	(See Generator output voltage (with i-ELOOP).)			• Generator • Related wiring harness
1AS	—	—	—		—	—
1AT	GND	Sensor shield	Ignition switched ON (engine off)		Below 1.0	• Related wiring harness
1AU	—	—	—		—	—
1AV	—	—	—		—	—
1AW	Purge control	Purge solenoid valve	(See Purge control.)			• Purge solenoid valve • Related wiring harness
1AX	—	—	—		—	—
1AY*2	Neutral switch No.2	Neutral switch No.2	Ignition switched ON (engine off)	Neutral	Below 1.0	• Neutral switch No.2 • Related wiring harness
				Except above	B+	
1AZ*2	Back-up light	Back-up light switch	Shift lever is at R position		Below 1.0	• Back-up light switch • Related wiring harness
			Shift lever is not at R position		B+	
1BA	—	—	—		—	—
1BB	—	—	—		—	—
1BC	IGT1	Ignition coil No.1	(See IGT1, IGT2, IGT3, IGT4 control.)			• Ignition coil No.1 • Related wiring harness
1BD	—	—	—		—	—
1BE	Oil pressure	Oil pressure switch	Ignition switched ON (engine off)		Below 1.0	• Oil pressure switch • Related wiring harness
			Idle (after warm up)		B+	

Terminal	Signal	Connected to	Test condition	Voltage (V)	inspection item
1BF*2	Neutral position	Neutral switch No.1	Shift lever is at neutral position	Below 1.0	• Neutral switch No.1 • Related wiring harness
			Shift lever is not at neutral position	B+	
1BG	—	—	—	—	—
1BH	GND	MAP sensor, IAT sensor No.2, fuel pressure sensor	Under any condition	Below 1.0	• Related wiring harness
1BI	IGT2	Ignition coil No.2	(See IGT1, IGT2, IGT3, IGT4 control.)		• Ignition coil No.2 • Related wiring harness
1BJ	—	—	—	—	—
1BK	—	—	—	—	—
1BL	—	—	—	—	—
1BM	Electric variable valve timing motor (rotation pulse)	Electric variable valve timing motor	(See Electric variable valve timing motor (rotation pulse) signal.)		• Electric variable valve timing motor • Related wiring harness
1BN	Electric variable valve timing motor (rotation direction)	Electric variable valve timing motor	(See Electric variable valve timing motor (rotation direction) signal.)		• Electric variable valve timing motor • Related wiring harness
1BO	IGT3	Ignition coil No.3	(See IGT1, IGT2, IGT3, IGT4 control.)		• Ignition coil No.3 • Related wiring harness
1BP*4	Generator field coil control	Generator	(See Generator field coil control signal (without i-ELOOP).)		• Generator • Related wiring harness
1BQ	—	—	—	—	—
1BR	GND	Sensor shield	Ignition switched ON (engine off)	Below 1.0	• Related wiring harness
1BS	—	—	—	—	—
1BT	Intake CMP	Intake CMP sensor	(See Intake CMP signal.)		• Intake CMP sensor • Related wiring harness
1BU	IGT4	Ignition coil No.4	(See IGT1, IGT2, IGT3, IGT4 control.)		• Ignition coil No.4 • Related wiring harness
1BV	Electric variable valve timing control	Electric variable valve timing driver	(See Electric variable valve timing control signal.)		• Electric variable valve timing driver • Related wiring harness
1BW	—	—	—	—	—
1BX	—	—	—	—	—
1BY	—	—	—	—	—
1BZ	Exhaust CMP	Exhaust CMP sensor	(See Exhaust CMP signal.)		• Exhaust CMP sensor • Related wiring harness
1CA	—	—	—	—	—
1CB	A/F sensor heater control	A/F sensor heater	(See A/F sensor heater control signal.)		• A/F sensor heater • Related wiring harness
1CC	—	—	—	—	—
1CD	—	—	—	—	—
1CE	—	—	—	—	—
1CF	—	—	—	—	—
1CG	—	—	—	—	—
1CH	—	—	—	—	—
1CI	—	—	—	—	—
1CJ	—	—	—	—	—
1CK	Engine oil control	Engine oil solenoid valve	(See Engine oil control signal.)		• Engine oil solenoid valve • Related wiring harness
1CL	—	—	—	—	—
1CM	—	—	—	—	—
1CN	—	—	—	—	—
1CO	Hydraulic variable valve timing control	OCV for hydraulic variable valve timing system	(See Hydraulic variable valve timing control signal.)		• OCV • Related wiring harness
1CP	—	—	—	—	—
1CQ	Drive-by-wire control (+)	Throttle valve actuator	(See Drive-by-wire control (+) signal.)		• Throttle valve actuator • Related wiring harness
1CR	Drive-by-wire control (–)	Throttle valve actuator	**Idle (after warm up)** Because the drive-by-wire control (-) terminal value varies depending on the vehicle, examination using only the ICQ terminal is not possible. When performing the inspection, perform it together with the ICR terminal. • Type A — B+ • Type B — Approx. 0		• Throttle valve actuator • Related wiring harness
1CS	—	—	—	—	—
1CT	—	—	—	—	—
1CU	Fuel injection control (+)	Fuel injector No.3	(See Fuel injection control (+) signal.)		• Fuel injector No.3 • Related wiring harness
1CV	Fuel injection control (–)	Fuel injector No.3	(See Fuel injection control (-) signal.)		• Fuel injector No.3

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
						• Related wiring harness
1CW	—	—	—		—	—
1CX	—	—	—		—	—
1CY	Fuel injection control (+)	Fuel injector No.2	(See Fuel injection control (+) signal.)			• Fuel injector No.2 • Related wiring harness
1CZ	Fuel injection control (–)	Fuel injector No.2	(See Fuel injection control (-) signal.)			• Fuel injector No.2 • Related wiring harness
1DA	—	—	—		—	—
1DB	—	—	—		—	—
1DC	Fuel injection control (+)	Fuel injector No.4	(See Fuel injection control (+) signal.)			• Fuel injector No.4 • Related wiring harness
1DD	Fuel injection control (–)	Fuel injector No.4	(See Fuel injection control (-) signal.)			• Fuel injector No.4 • Related wiring harness
1DE	—	—	—		—	—
1DF	—	—	—		—	—
1DG	Fuel injection control (+)	Fuel injector No.1	(See Fuel injection control (+) signal.)			• Fuel injector No.1 • Related wiring harness
1DH	Fuel injection control (–)	Fuel injector No.1	(See Fuel injection control (-) signal.)			• Fuel injector No.1 • Related wiring harness
1DI	High pressure fuel pump control (+)	High pressure fuel pump	(See High pressure fuel pump control (+) signal.)			• High pressure fuel pump • Related wiring harness
1DJ	High pressure fuel pump control (–)	High pressure fuel pump	(See High pressure fuel pump control (-) signal.)			• High pressure fuel pump • Related wiring harness
2A	Battery voltage	Battery	Ignition switched ON (engine off)		B+	• Battery • Related wiring harness
2B	Main relay control	Main relay	Ignition switched ON (engine off)		Below 1.0	• Main relay • Related wiring harness
2C*3	DC-DC converter (i-ELOOP) control	DC-DC converter (i-ELOOP)	Idle (after warm up)		0 or B+	• DC-DC converter (i-ELOOP) • Related wiring harness
2D	—	—	—		—	—
2E	—	—	—		—	—
2F	Ignition (IG1)	IG1 relay	Ignition switched ON (engine off)		B+	• IG1 relay • Related wiring harness
2G	—	—	—		—	—
2H	Starter cut-off control	Starter relay, start stop unit	Ignition switched ON (engine off)	Selector lever position is not P or N position	B+	• Starter relay • Start stop unit • Related wiring harness
				Selector lever position is P or N position	Below 1.0	
2I	Cooling fan control	Fan control module	(See Cooling fan control signal.)			• Fan control module • Related wiring harness
2J	—	—	—		—	—
2K	—	—	—		—	—
2L	—	—	—		—	—
2M	—	—	—		—	—
2N	—	—	—		—	—
2O	—	—	—		—	—
2P	Brake (No.2)	Brake switch (No.2 signal)	Brake pedal released		Below 1.0	• Brake switch (No.2 signal) • Related wiring harness
			Brake pedal depressed		B+	
2Q	—	—	—		—	—
2R	Ambient temperature	Ambient temperature sensor	Ignition switched ON (engine off)	AAT 20 °C {68 °F}	Approx. 2.70	• Ambient temperature sensor • Related wiring harness
				AAT 30 °C {104 °F}	Approx. 1.80	
2S	CAN_H	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
2T	CAN_L	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
2U	—	—	—		—	—
2V	—	—	—		—	—
2W	—	—	—		—	—
2X	Power brake unit vacuum	Vacuum sensor 2	Ignition switched ON (engine off)	Brake pedal depressed (10 times)	Approx. 4.01	• Vacuum sensor 2 • Related wiring harness
			Idle (after warm up)	Brake pedal released	Approx. 0.36—0.40	
2Y	—	—	—		—	—
2Z	—	—	—		—	—
2AA	—	—	—		—	—
2AB	Brake (No.1)	Brake switch (No.1 signal)	Brake pedal released		Below 1.0	• Brake switch (No.1 signal) • Related wiring harness
			Brake pedal depressed		B+	
2AC	—	—	—		—	—
2AD	—	—	—		—	—

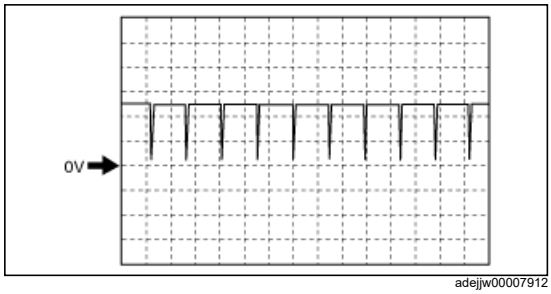
Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
2AE	—	—	—		—	—
2AF	A/C cut-off control	A/C relay	A/C relay OFF		B+	• A/C relay • Related wiring harness
			A/C relay ON		Below 1.0	
2AG	—	—	—		—	—
2AH*2	CPP	CPP switch, start stop unit	Clutch pedal fully depressed		Below 1.0	• CPP switch • Start stop unit • Related wiring harness
			Clutch pedal released		B+	
2AI	GND	MAF sensor, IAT sensor No.1, refrigerant pressure sensor, ambient temperature sensor, clutch stroke sensor*2, vacuum sensor 2	Under any condition		Below 1.0	• Related wiring harness
2AJ	—	—	—		—	—
2AK	DC-DC converter control*4	DC-DC converter	Ignition switched ON (engine off)		Below 1.0	• DC-DC converter • Related wiring harness
	DC-DC converter (i-ELOOP) control*3	DC-DC converter (i-ELOOP)	Ignition switched ON (engine off)		Below 1.0	• DC-DC converter (i-ELOOP) • Related wiring harness
2AL	—	—	—		—	—
2AM	Cooling fan control	Cooling fan relay	Cooling fan operating		Below 1.0	• Cooling fan relay • Related wiring harness
			Cooling fan not operating		B+	
2AN	—	—	—		—	—
2AO	Constant voltage (Vref)	APP sensor No.2	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
2AP*2	Clutch stroke sensor	Clutch stroke sensor	Ignition switched ON (engine off)	Clutch pedal released	Approx. 4.5	• Clutch stroke sensor • Related wiring harness
				Clutch pedal fully depressed	Below 1.0	
2AQ	—	—	—		—	—
2AR	—	—	—		—	—
2AS	—	—	—		—	—
2AT	Constant voltage (Vref)	MAF sensor, IAT sensor No.1	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
2AU	Constant voltage (Vref)	Vacuum sensor 2, clutch stroke sensor*2	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
2AV	Refrigerant pressure	Refrigerant pressure sensor	Refrigerant pressure: 1.0 MPa {10 kgf/cm^2, 145 psi}		Approx. 1.58	• Refrigerant pressure sensor • Related wiring harness
			Refrigerant pressure: 1.1 MPa {11 kgf/cm^2, 160 psi}		Approx. 1.72	
			Refrigerant pressure: 1.2 MPa {12 kgf/cm^2, 174 psi}		Approx. 1.88	
2AW	Fuel pump control	Fuel pump relay	Ignition switched ON (engine off)		B+	• Fuel pump relay • Related wiring harness
			Idle (after warm up)		Below 1.0	
2AX	—	—	—		—	—
2AY	—	—	—		—	—
2AZ	Constant voltage (Vref)	Refrigerant pressure sensor	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
2BA	Constant voltage (Vref)	APP sensor No.1	Ignition switched ON (engine off)		Approx. 5.0	• Related wiring harness
2BB	APP (No.2)	APP sensor No.2	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 0.58	• APP sensor No.2 • Related wiring harness
				Accelerator pedal depressed	Approx. 2.30	
2BC	—	—	—		—	—
2BD	Selector lever position*1	TCM, start stop unit	Selector lever position is not P or N position		B+	• TCM • Start stop unit • Related wiring harness
			Selector lever position is P or N position		Below 1.0	
	Starter interlock*2	Starter interlock switch, start stop unit	Clutch pedal fully depressed		Below 1.0	• Starter interlock switch • Start stop unit • Related wiring harness
			Clutch pedal released		B+	
2BE	Intake air	MAF sensor, IAT sensor No.1	Because this terminal is for SENT signal, integrity determination by terminal voltage is not possible.			• MAF sensor, IAT sensor No.1 • Related wiring harness
2BF	—	—	—		—	—
2BG	GND	APP sensor No.2	Under any condition		Below 1.0	• Related wiring harness
2BH	GND	APP sensor No.1	Under any condition		Below 1.0	• Related wiring harness
2BI	—	—	—		—	—
2BJ	HO2S heater control	HO2S heater	(See HO2S heater control signal.)			• HO2S heater • Related wiring harness
2BK	HO2S (+)	HO2S	Idle (after warm up)		Approx. 2.31	• HO2S • Related wiring harness
2BL	HO2S (–)	HO2S	Idle (after warm up)		Approx. 2.05	• HO2S • Related wiring harness
2BM	GND	Sensor shield	Ignition switched ON (engine off)		Below 1.0	• Related wiring harness
2BN	APP (No.1)	APP sensor No.1	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 0.79	• APP sensor No.1 • Related wiring harness
				Accelerator pedal depressed	Approx. 4.56	
2BO	Main power supply	Main relay	Ignition switched ON (engine off)		B+	• Main relay

Terminal	Signal	Connected to	Test condition	Voltage (V)	inspection item
					• Related wiring harness
2BP	—	—	—	—	—
2BQ	GND	GND	Ignition switched ON (engine off)	Below 1.0	• Related wiring harness
2BR	Main power supply	Main relay	Ignition switched ON (engine off)	B+	• Main relay • Related wiring harness
2BS	—	—	—	—	—
2BT	GND	GND	Ignition switched ON (engine off)	Below 1.0	• Related wiring harness

*1 :AT
*2 :MT
*3 :With i-ELOOP
*4 :Without i-ELOOP

Inspection Using An Oscilloscope (Reference)

CKP signal



adejjw00007912

PCM terminals

- 1F(+)—body ground(–)

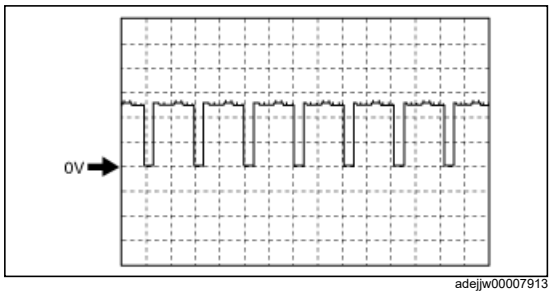
Oscilloscope setting

- 2 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up)

Electric variable valve timing driver (diagnostic) signal



adejjw00007913

PCM terminals

- 1AD(+)—body ground(–)

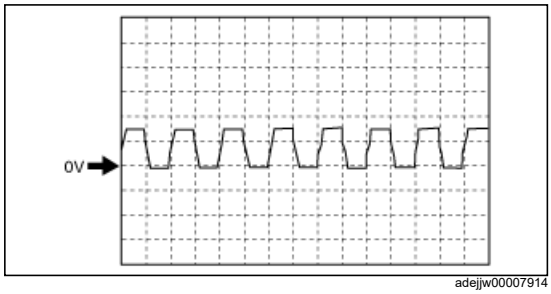
Oscilloscope setting

- 2 V/DIV (Y), 100 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up)

Generator output voltage (without i-ELOOP)



adejjw00007914

PCM terminals

- 1AG(+)—body ground(–)

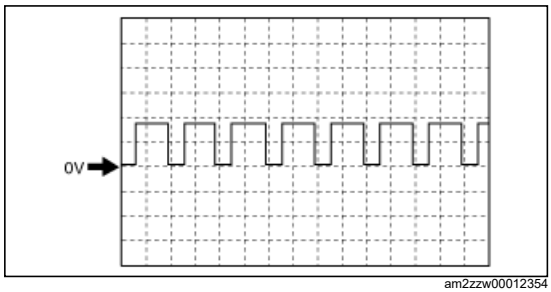
Oscilloscope setting

- 5 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Generator field coil control signal (with i-ELOOP)



am2zzw00012354

PCM terminals

- 1AQ(+)—body ground(–)

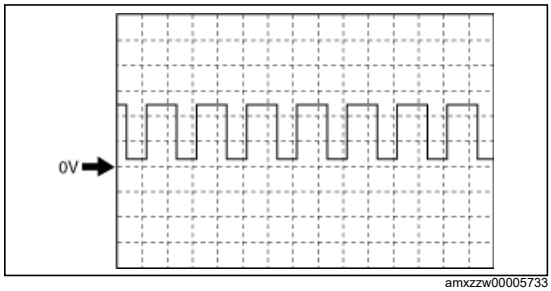
Oscilloscope setting

- 5 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Generator output voltage (with i-ELOOP)



amxzzw00005733

PCM terminals

- 1AR(+)—body ground(–)

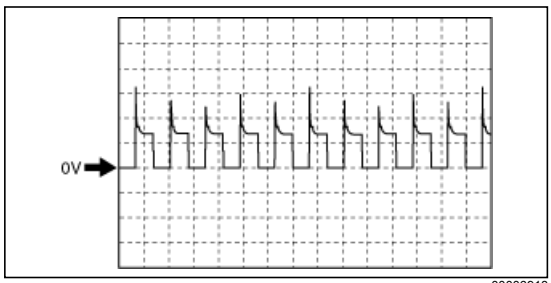
Oscilloscope setting

- 2 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Purge control



amxuuw00006918

PCM terminals

- 1AW(+)—body ground(–)

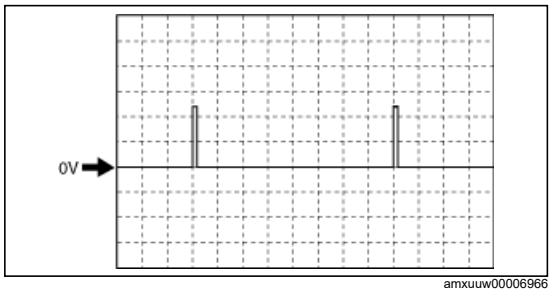
Oscilloscope setting

- 10 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition

- Racing (Engine speed: 2,000 rpm)

IGT1, IGT2, IGT3, IGT4 control



amxuuw00006966

PCM terminals

- IGT1 (ignition coil No.1): 1BC(+)—body ground(–)
- IGT2 (ignition coil No.2): 1BI(+)—body ground(–)
- IGT3 (ignition coil No.3): 1BO(+)—body ground(–)
- IGT4 (ignition coil No.4): 1BU(+)—body ground(–)

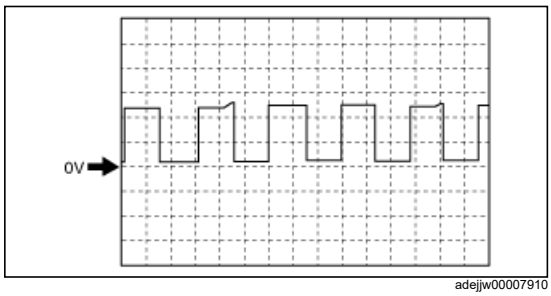
Oscilloscope setting

- 2 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Electric variable valve timing motor (rotation pulse) signal



adejjw00007910

PCM terminals

- 1BM(+)—body ground(–)

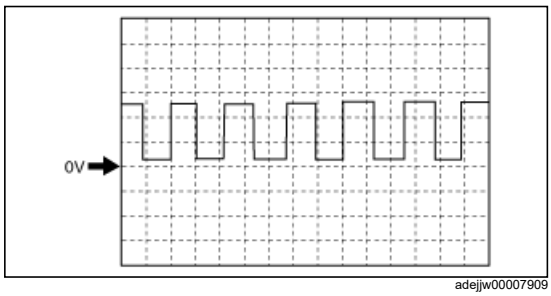
Oscilloscope setting

- 2 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up)

Electric variable valve timing motor (rotation direction) signal



adejjw00007909

PCM terminals

- 1BN(+)—body ground(–)

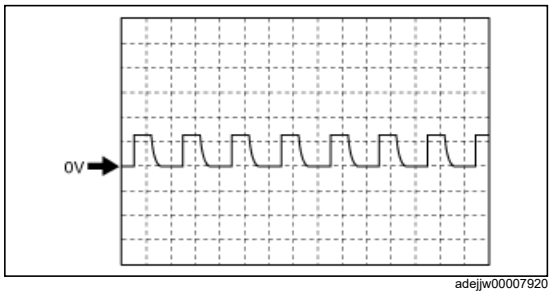
Oscilloscope setting

- 2 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up)

Generator field coil control signal (without i-ELOOP)



adejjw00007920

PCM terminals

- 1BP(+)—body ground(–)

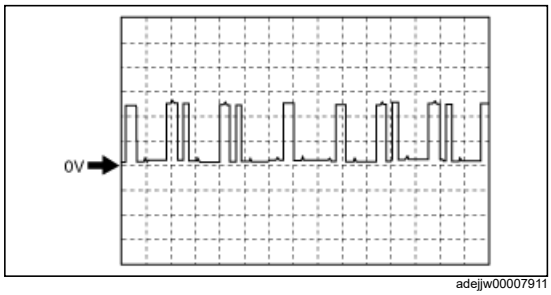
Oscilloscope setting

- 1 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Intake CMP signal



adejjw00007911

PCM terminals

- 1BT(+)—body ground(–)

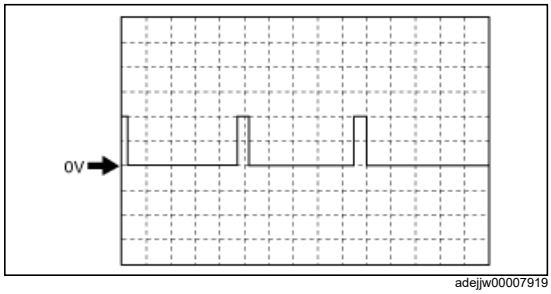
Oscilloscope setting

- 2 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up)

Electric variable valve timing control signal



adejjw00007919

PCM terminals

- 1BV(+)—body ground(–)

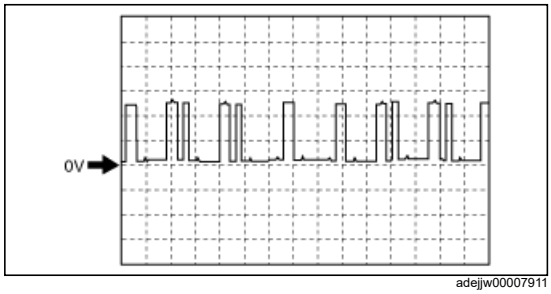
Oscilloscope setting

- 2 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Exhaust CMP signal



adejjw00007911

PCM terminals

- 1BZ(+)—body ground(–)

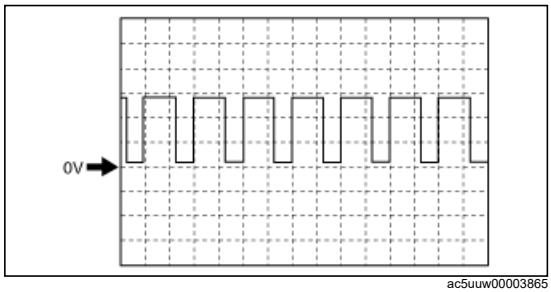
Oscilloscope setting

- 2 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up)

A/F sensor heater control signal



ac5uuw00003865

PCM terminals

- 1CB(+)—body ground(–)

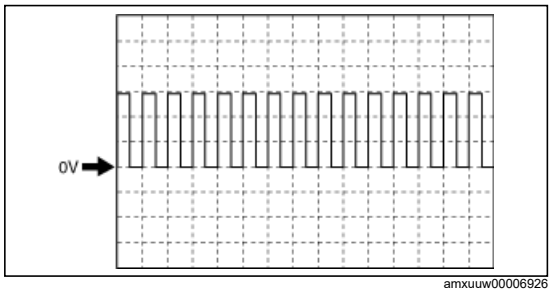
Oscilloscope setting

- 5 V/DIV (Y), 10 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Engine oil control signal



amxuuw00006926

PCM terminals

- 1CK(+)—body ground(–)

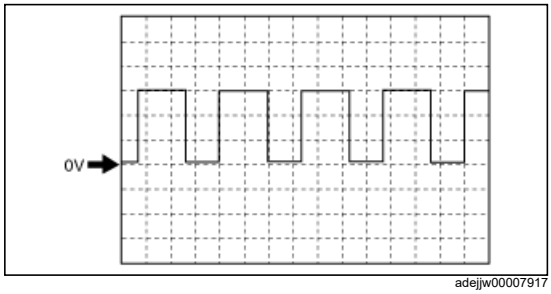
Oscilloscope setting

- 5 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Hydraulic variable valve timing control signal



adejjw00007917

PCM terminals

- 1CO(+)—body ground(–)

Oscilloscope setting

- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition

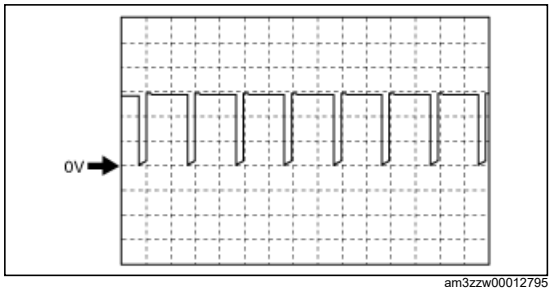
- Racing (Engine speed: 2,000 rpm)

Drive-by-wire control (+) signal

Note

- The wave pattern is either type A or B depending on the PCM specifications.

Type A



am3zzw00012795

PCM terminals

- 1CQ(+)—1CR(–)

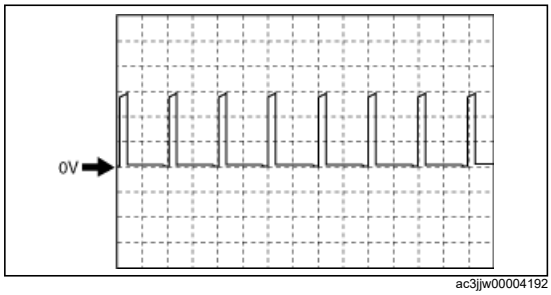
Oscilloscope setting

- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Type B



ac3jjw00004192

PCM terminals

- 1CQ(+)—1CR(–)

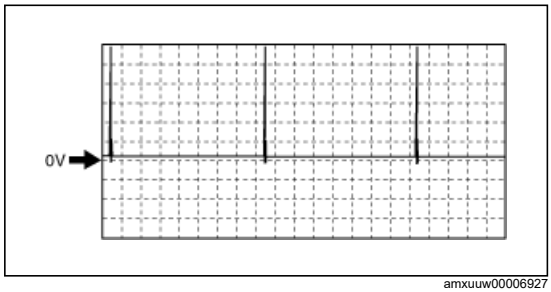
Oscilloscope setting

- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Fuel injection control (+) signal



amxuuw00006927

PCM terminals

- Fuel Injection No.1: 1DG(+)—body ground(–)
- Fuel Injection No.2: 1CY(+)—body ground(–)
- Fuel Injection No.3: 1CU(+)—body ground(–)
- Fuel Injection No.4: 1DC(+)—body ground(–)

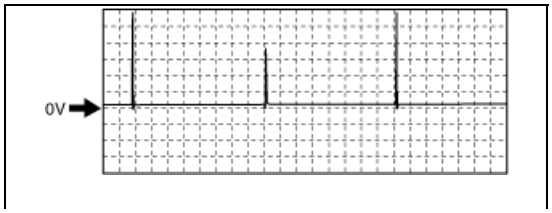
Oscilloscope setting

- 10 V/DIV (Y), 10 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Fuel injection control (-) signal



amxuuw00006919

PCM terminals

- Fuel Injection No.1: 1DH(+)—body ground(–)
- Fuel Injection No.2: 1CZ(+)—body ground(–)
- Fuel Injection No.3: 1CV(+)—body ground(–)
- Fuel Injection No.4: 1DD(+)—body ground(–)

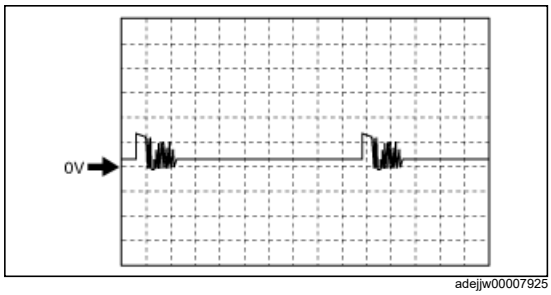
Oscilloscope setting

- 10 V/DIV (Y), 10 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

High pressure fuel pump control (+) signal



adejjw00007925

PCM terminals

- 1DI(+)—body ground(–)

Oscilloscope setting

- 10 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

High pressure fuel pump control (-) signal

amxuuw00006920

PCM terminals

- 1DJ(+)—body ground(–)

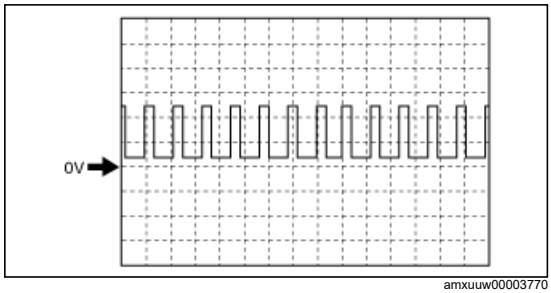
Oscilloscope setting

- 10 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition

- Idle after warm up

Cooling fan control signal



amxuuw00003770

PCM terminals

- 2I(+)—body ground(–)

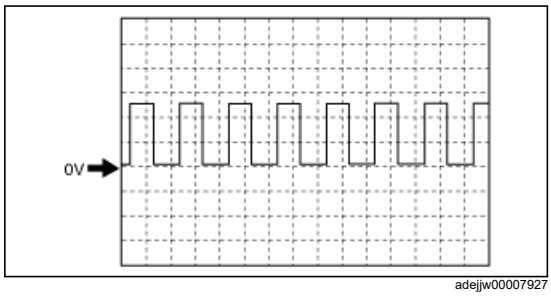
Oscilloscope setting

- 5 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition

- Idle (after warm up and no load)

HO2S heater control signal



adejjw00007927

PCM terminals

- 2BJ(+)—body ground(–)

Oscilloscope setting

- 5 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition

- Idle (immediately after starting engine)

Using The M-MDS

Note

- PIDs for the following parts are not available on this model. Go to the appropriate part inspection page.
  - Intake CMP sensor and exhaust CMP sensor (See CAMSHAFT POSITION (CMP) SENSOR INSPECTION [SKYACTIV-G 1.5, SKYACTIV-G 2.0].)
  - Main relay (See RELAY INSPECTION.)

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Measure the PID value.
   - If PID value is not within the specification, follow the instructions in action column.

Note

- The PID/DATA MONITOR function monitors the calculated value of the input/output signals in the PCM. Therefore, an output device malfunction is not directly indicated as a malfunction of the monitored value for the output device. If a monitored value of an output device is out of specification, inspect the monitored value of the input device related to the output control.
- The simulation items that are used in the engine control system operation inspection are as follows.
  - ACCS, ARPMDES, EVAPCP, FAN1, FAN2, FP, INJ_1, INJ_2, INJ_3, INJ_4, OIL_P_SOL

—: Not applicable

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
AAT	°C, °F	• Ambient air temperature	Calculation	• Displays ambient air temperature	2AI
ABV_OIL_P	Displays in the M-MDS but it does not operate.				
AC_PRES	KPa {MPa}, mBar {Bar}, psi, in H20	Refrigerant pressure input from refrigerant pressure sensor	Calculation	• Displays refrigerant pressure	2AV
	V	Refrigerant pressure sensor voltage	Input	• Refrigerant pressure is 449 kPa {4.58 kgf/cm^2, 65.1 psi}: Approx. 0.76 V • Refrigerant pressure is 729 kPa {7.43 kgf/cm^2, 106 psi}: Approx. 1.18 V	
AC_REQ	Off/On	A/C switch status received by PCM via CAN	Input	• A/C switch off: Off • A/C switch on: On	CAN (2S, 2T)
ACCS	Off/On	A/C relay status input from A/C relay	Input	• A/C relay off: Off • A/C relay on: On	2AF
AIRSHUT_ACT	Displays in the M-MDS but it does not operate.				
AIRSHUT_CAL	Displays in the M-MDS but it does not operate.				
ALTF	%	Field coil current control signal output to generator	Calculation	• Ignition switched ON (engine off): 0% • Idle: Approx. 41—54% • Racing (Engine speed is 2,000 rpm): Approx. 24—34%	1AQ*3, 1BP*4
ALTF_ACT*3	%	Actually measured value of field coil current signal input from generator	Calculation	• Displays actual generator field current control duty value	—
ALTT V*4	V	Generator output voltage	Input	• Idle (no E/L): Approx. 14 V (This is internal calculation value and differs from terminal voltage)	1AG
AMB_TEMP	°C, °F	Actually measured ambient temperature input from ambient temperature sensor	Calculation	• Displays ambient air temperature	2R
APP	%	Accelerator pedal opening angle (relative value) with the fully released status as 0% and fully depressed status as 100%	Calculation	• Accelerator pedal released: Approx. 0% • Accelerator pedal fully depressed: Approx. 100%	—
APP1	%	Accelerator pedal opening angle (absolute value) input from APP sensor No.1	Calculation	• Accelerator pedal released: Approx. 16% • Accelerator pedal fully depressed: Approx. 91%	2BN
	V	APP sensor No.1 voltage	Input	• Accelerator pedal released: Approx. 0.78 V • Accelerator pedal fully depressed: Approx. 4.54 V	
APP2	%	Accelerator pedal opening angle (absolute value) input from APP sensor No.2	Calculation	• Accelerator pedal released: Approx. 7.84%	2BB

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
				• Accelerator pedal fully depressed: Approx. 45.49%	
	V	APP sensor No.2 voltage	Input	• Accelerator pedal released: Approx. 0.39 V • Accelerator pedal fully depressed: Approx. 2.27 V	
ARPMDES	RPM	Target engine speed	Calculation	• Displays target engine speed	—
AXIO_G_S	Off/On	Recommended Gear Supported	Calculation	• Displays recommended Gear Supported	—
AXIO_TG_V	Off/On	Recommended transmission gear for current vehicle conditions	Calculation	• Displays recommended transmission gear for current vehicle conditions	—
BARO	KPa {MPa}, mBar {Bar}, psi, in H20	Actually measured barometric pressure input from barometric pressure sensor built into PCM	Calculation	• Displays BARO	—
BATT_CUR	A	Battery charge/discharge current	Calculation	• Displays battery charge/discharge current value	CAN (2S, 2T)
BATT_DAY	—	Number of days elapsed since current sensor initialization	Calculation	• Displays vehicle battery days in service	CAN (2S, 2T)
BATT_RES	—	Battery internal resistance (estimated)	Calculation	• Displays battery inferred internal resistance	CAN (2S, 2T)
BATT_SOC	%	Battery charge condition (estimated)	Calculation	• Displays battery estimated state of charge	CAN (2S, 2T)
BATT_TEMP	°C, °F	Battery temperature	Calculation	• Displays battery fluid temperature	CAN (2S, 2T)
BATT_V	V	Battery voltage	Calculation	• Displays battery voltage	CAN (2S, 2T)
BBP	KPa {MPa}, mBar {Bar}, psi, in H20	Power brake unit vacuum input from vacuum sensor 2	Calculation	• Displays power brake unit vacuum	2X
	V	Vacuum sensor 2 voltage	Input	**Ignition switched ON** • Power brake unit vacuum is 7.96 kPa {0.0812 kgf/cm^2, 1.15 psi}: approx. 0.31 V • Power brake unit vacuum is 97.23 kPa {0.9915 kgf/cm^2, 14.10 psi}: approx. 3.93 V	
BFP	KPa {MPa}, mBar {Bar}, psi, in H20	Actually measured brake fluid pressure input from brake fluid pressure sensor built into DSC HU/CM via CAN	Calculation	• Displays brake fluid pressure	CAN (2S, 2T)
BOO	High/Low	Brake switch (No.1 signal) input status	Calculation	• Brake pedal released: Low • Brake pedal fully depressed: High	2AB
BPA	High/Low	Brake switch (No.2 signal) input status	Calculation	• Brake pedal released: Low • Brake pedal fully depressed: High	2P
BT_AH_ACCMLTD	—	Battery accumulated amp hour	Calculation	• Displays the battery accumulated amp hour	—
CATT11_DSD	°C, °F	Estimated catalytic converter temperature	Calculation	• Displays estimated catalytic converter temperature **Idle (after warm up)** — ECT is 88 °C {190 °F}: Approx. 304 °C {579 °F}	—
CHRGLP	Off/On	Charging system warning light illumination status	Calculation	• Charging system warning light not illuminated: Off • Charging system warning light illuminated: On	—
CL_NMHC_MN_CP	Displays in the M-MDS but it does not operate.				
CLR_DIST	—	—	Calculation	• Displays the mileage after DTC cleared	—
CLU_CUT_SW*1	Off/On	Starter interlock switch status	Calculation	• Clutch pedal released: Off (starter interlock switch OFF) • Clutch pedal fully depressed: On (starter interlock switch ON)	2BD
CLU_SW*1	Off/On	CPP switch status	Calculation	• Displays CPP switch status	2AH
CMP_MNT_ENA	Displays in the M-MDS but it does not operate.				
COMP3_MNT	—	Comprehensive component monitoring completed	Calculation	• Displays the comprehensive component monitoring completed.	—
CP_IGN_SP	Yes/No	Compression ignition monitoring supported	Calculation	• Displays the compression ignition monitoring supported.	—
CPP*1	Off/On	Clutch pedal position	Calculation	• Clutch pedal released: Off • Clutch pedal fully depressed: On	2AH
CPP*1	%	Clutch pedal position input from clutch stroke sensor	Calculation	• Clutch pedal released: approx. 0.2% • Clutch pedal fully depressed: approx. 93.5%	2AP
CPP/PNP*1	Off/On	Shift lever position	Calculation	• Other than neutral: Off • Neutral: On	1BF, 1AY
ECT	°C, °F	Engine coolant temperature input from ECT sensor	Calculation	• Displays ECT	1AH, 1AM
	V	ECT sensor voltage	Input	**Ignition switched ON (engine off)** • ECT is 29 °C {84 °F}: Approx. 2.65 V **Idle (after warm up)** • ECT is 88 °C {190 °F}: Approx. 0.71 V	
ENG_CL_V_POS	%	Coolant control valve opening angle	Calculation	• Displays target coolant control valve opening angle	—
ENG_CL_V_POS_R	%	Coolant control valve opening angle raw	Input	• Ignition switched ON (engine off): Approx. 63.92 %	1I
EOL	mm, in	Engine oil level	Converted value	• Displays in the M-MDS but it does not operate.	—
EOP	KPa {MPa}, mBar {Bar}, psi, in H20	Engine oil pressure	Converted value	• Switch ignition ON (engine off): 0—1 kPa {0.00—0.01 kgf/cm2, 0.0—0.1 psi}	1AS

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
				• Idle: Approx. 150 kPa {1.53 kgf/cm2, 21.8 psi}	
EOT2	°C, °F	Engine oil temperature from engine oil level sensor	Converted value	• Displays in the M-MDS but it does not operate.	—
EQ_RAT11	—	Excess air factor (estimated value) to theoretical air/fuel ratio (14.7) by fuel feedback control	Calculation	**Idle (after warm up)** • ECT is 89 °C {192 °F}: Approx. 1	—
EQ_RAT11_DSD	—	Target excess air factor (estimated value) to theoretical air/fuel ratio (14.7) by fuel feedback control	Calculation	• Indicate target lambda (Excess air factor = supplied air amount / theoretical air/fuel ratio)	—
ETC_ACT	°	Actual throttle valve opening angle	Calculation	**Ignition switched ON (engine off)** • Accelerator pedal released: Approx. 12.89 ° • Accelerator pedal fully depressed: Approx. 86.03 ° **Idle (after warm up)** • Accelerator pedal released: Approx. 2.77—2.84 ° (ECT is 89 °C {192 °F})	—
ETC_DSD	°	Target throttle valve opening angle	Calculation	• Displays target TP opening angle **Ignition switched ON (engine off)** — Accelerator pedal released: Approx. 12.9 ° — Accelerator pedal fully depressed: Approx. 86 °	—
	%	Target throttle valve opening angle (percent)	Calculation	• Displays target TP opening angle (percent) **Ignition switched ON (engine off)** — Accelerator pedal released: Approx. 14.9% — Accelerator pedal fully depressed: Approx. 100 %	
EVAP_SYS_MN	Yes/No	Evaporative system monitoring	Calculation	• Displays the evaporative system monitoring.	—
EVAP_SYS_MN_CP	Yes/No	Evaporative system monitoring completed	Calculation	• Displays the evaporative system monitoring completed.	—
EVAPCP	%	Purge solenoid valve control duty value	Calculation	• Idle (after warm up): 0% (Engine coolant temperature 59 °C {140 °F} or less) • Racing (Engine speed 2,000 rpm): 33—35% (ECT is 94 °C {201 °F}) • Racing (Engine speed 4,000 rpm): 53—55% (ECT is 94 °C {201 °F})	1AW
EVAPCV	Displays in the M-MDS but it does not operate.				
FAN_DUTY	Displays in the M-MDS but it does not operate.				
FAN1	Off/On	Cooling fan relay operation status	Calculation	• Cooling fan relay off: Off • Cooling fan relay on: On	2AM
FAN2	Displays in the M-MDS but it does not operate.				
FAN3	Displays in the M-MDS but it does not operate.				
FCL	Displays in the M-MDS but it does not operate.				
FEO_ICC_LT	—	Fueled Engine Operation Ignition Cycle Counter (Lifetime)	Calculation	• Displays the fueled Engine Operation Ignition Cycle Counter (Lifetime).	—
FEO_ICC_R	—	Fueled Engine Operation Ignition Cycle Counter (Recent)	Calculation	• Displays the fueled Engine Operation Ignition Cycle Counter (Recent).	—
FIA	—	Fuel injection amount	Calculation	• Displays fuel injection amount **Idle (after warm up)** — ECT is 90 °C {194 °F}: Approx. 6	—
FLI	%	Fuel level	Calculation	• Fuel gauge level F: Approx. 100% • Fuel gauge level E: Approx. 0%	CAN (2S, 2T)
FP	Off/On	Fuel pump relay operation status	Calculation	• Ignition switched ON (engine off): Off • Cranking: On • Idle (after warm up): On	2AW
FP_DUTY	Displays in the M-MDS but it does not operate.				
FRP_A	KPa {MPa}, mBar {Bar}, psi, in H20	Actual fuel distributor pressure	Calculation	• Ignition switched ON (engine off): Approx. 3.36 MPa {34.3 kgf/cm^2, 487 psi} • Idle (after warm up): Approx. 2.96—3.13 MPa {30.2—31.9 kgf/cm^2, 430—453 psi} • Racing (engine speed is 2,000 rpm): Approx. 3.3 MPa {34 kgf/cm^2, 479 psi} • Racing (engine speed is 4,000 rpm): Approx. 4.94 MPa {50.4 kgf/cm^2, 716 psi}	—
FRP_A_CMD	KPa {MPa}, mBar {Bar}, psi, in H20	Target fuel distributor pressure	Calculation	• Ignition switched ON (engine off): Approx. 6 MPa {61 kgf/cm^2, 870 psi} • Idle (after warm up): Approx. 3.1 MPa {32 kgf/cm^2, 450 psi} • Racing (engine speed is 2,000 rpm): Approx. 3.37 MPa {34.4 kgf/cm^2, 489 psi} • Racing (engine speed is 4,000 rpm): Approx. 4.97 MPa {50.7 kgf/cm^2, 721 psi}	—

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
FRP_A_CMD_S	No/Yes	Presence/non-presence of target fuel distributor pressure	Calculation	• With fuel distributor: Yes • Without fuel distributor: No	—
FRP_A_S	No/Yes	Presence/non-presence of actual fuel distributor	Calculation	• With fuel distributor: Yes • Without fuel distributor: No	—
FRP_GAUGE	No/Yes	Fuel rail pressure (gauge)	Calculation	• Displays fuel rail pressure (gauge)	—
FRT_A	°C, °F	Fuel distributor temperature	Calculation	• Displays fuel distributor temperature	—
FT_A_S	No/Yes	Presence/non-presence of fuel distributor temperature	Calculation	• With fuel distributor: Yes • Without fuel distributor: No	—
FTP	Displays in the M-MDS but it does not operate				
FU_SYS_CP	No/Yes	Fuel system monitoring completed	Calculation	• Displays fuel system monitoring completed	—
FU_SYS_ENA	No/Yes	Fuel system monitoring enabled	Calculation	• Displays fuel system monitoring enabled	—
FUEL_LO_P	Displays in the M-MDS but it does not operate				
FUEL_LO_T	Displays in the M-MDS but it does not operate				
FUEL_P_DSD	KPa {MPa}, mBar {Bar}, psi, in H20	Target fuel pressure (high pressure fuel pump)	Calculation	• Displays target fuel pressure (high pressure fuel pump)	—
FUEL_PRES	KPa {MPa}, mBar {Bar}, psi, in H20	Fuel pressure input from fuel pressure sensor	Calculation	• Displays fuel pressure	1W
	V	Fuel pressure sensor voltage	Input	Idle (ECT 80 °C {176 °F}) • Fuel pressure is 10 MPa {102 kgf/cm^2, 1450 psi}: Approx. 1.4 V	
FUELPW	Sec	Fuel injection pulse width (fuel injector energization time) output to fuel injector	Calculation	• Idle (after warm up): Approx. 1.71 ms • Racing (engine speed is 2,000 rpm): Approx. 1.35 ms • Racing (engine speed is 4,000 rpm): Approx. 1.21 ms	No.1:1DG/1DH No.2:1CY/1CZ No.3:1CU/1CV No.4:1DC/1DD
FUELSYS	OL/CL/ OL_Drive/ OL_Fault/ CL_Fault	Feedback status of fuel injection control is displayed OL: Feedback control is disabled at cold engine start CL: During feedback control by A/F sensor and HO2S OL_Drive: While feedback control is stopped OL_Fault: Feedback control is disable due to system malfunction CL_Fault: During feedback control with either A/F sensor or HO2S having a malfunction	Calculation	• Idle (after warm up): OL or CL • Racing (engine speed is 2,000 rpm): CL • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): OL-Drive	—
GEAR*2	Unknown/1st/2nd/3rd/4th/5th/6th/Not in P/Park/Neutral/Drive/Reverse	Gear commanded	Calculation	• Selector lever at P position: Park • Selector lever at R position: Reverse • Selector lever at N position: Neutral • Selector lever is in D or M position while vehicle is stopped: 1st	CAN (2S, 2T)
GENVDSD	V	Generator voltage desired	Calculation	• Displays generator voltage desired.	—
HTR11	Off/On	A/F sensor heater operation status	Calculation	• Ignition switched ON (engine off): Off • Idle (after warm up): On	1CB
	%	A/F sensor heater control duty value	Calculation	• Ignition switched ON (engine off): 0% • Idle (after warm up): 32—34%	
HTR12	Off/On	HO2S heater operation status	Calculation	• Ignition switched ON (engine off): Off • Idle (after warm up): On	2BJ
	%	HO2S heater control duty value	Calculation	• Ignition switched ON (engine off): 0% • Idle (after warm up): 41—48%	
IAT	°C, °F	Intake air temperature (No.1) input from IAT sensor No.1	Calculation	• Displays IAT (No.1)	2BE
	V	IAT sensor No.1 voltage	Input	• IAT is 20 °C {68 °F}: Approx. 2.70 V • IAT is 40 °C {104 °F}: Approx. 1.80 V • IAT is 60 °C {140 °F}: Approx. 1.20 V	
IAT2	°C, °F	Intake air temperature (No.2) input from IAT sensor No.2	Calculation	• Displays IAT (No.2)	1Q
	V	IAT sensor No.2 voltage	Input	• IAT2 is 20 °C {68 °F}: Approx. 3.57 V • IAT2 is 40 °C {104 °F}: Approx. 2.70 V • IAT2 is 60 °C {140 °F}: Approx. 1.87 V	
IAT_B1S1	°C, °F	Intake air temperature (No.1) input from IAT sensor No.1	Calculation	• Displays IAT (No.1)	2BE
	V	IAT sensor No.1 voltage	Input	• IAT is 20 °C {68 °F}: Approx. 2.70 V • IAT is 40 °C {104 °F}: Approx. 1.80 V • IAT is 60 °C {140 °F}: Approx. 1.20 V	
IAT_B1S2*5	°C, °F	Intake air temperature (No.2) input from IAT sensor No.2	Calculation	• Displays IAT (No.2)	1Q
	V	IAT sensor No.2 voltage	Input	• IAT2 is 20 °C {68 °F}: Approx. 3.57 V • IAT2 is 40 °C {104 °F}: Approx. 2.70 V • IAT2 is 60 °C {140 °F}: Approx. 1.87 V	
IG_CNT_LT	—	Ignition Counter (Lifetime)	Calculation	• Displays the ignition Counter (Lifetime).	—
IG_CNT_R	—	Ignition Counter (Recent)	Calculation	• Displays the ignition Counter (Recent).	—

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
INGEAR	Off/On	Gears are engaged	Calculation	**MTX** • When the following conditions are satisfied: On — Other than neutral — Clutch pedal released • Except above: Off **ATX** • Selector lever is in R, D, or M position: On • Except above: Off	CAN (2S, 2T)
INLET_PRES	Displays in the M-MDS but it does not operate.				
INT_CO_TEMP	Displays in the M-MDS but it does not operate.				
I-Stop_OFF	Off/On	i-stop OFF mode	Calculation	• i-stop OFF switch OFF: Off • i-stop OFF switch ON: On	CAN (2S, 2T)
I-Stop_TRD	Off/On	i-stop transmission D position selected status	Calculation	• D position:On • Except above: Off	CAN (2S, 2T)
I-Stop_VSP	Off/On	i-stop vehicle speed history flag	Calculation	• Vehicle speed in which engine stop condition is met via i-stop control is detected: On • Except above: Off	CAN (2S, 2T)
I-Stop_VST	Off/On	i-stop vehicle stop flag	Calculation	• Vehicle stop predicted: On • Except above: Off	CAN (2S, 2T)
ISV_DSD	Displays in the M-MDS but it does not operate.				
IVS	Off Idle/Idle	Idle flag	Calculation	• Racing: Off Idle • Idle: Idle	—
KNOCKR	°	Ignition timing correction for suppressing engine knock (Performs retard correction (negative indication) according to the occurrence of engine knock, and it approaches approx. 0° by the advance correction due to engine knock suppression.)	Calculation	• Ignition switched ON (engine off): 0 ° • Idle (after warm up): 0 °	—
LOAD	%	Ratio of actual amount of intake air to the maximum air charging amount (mass volume) of cylinder	Calculation	• Idle (after warm up): Approx. 22.35% • Racing (engine speed is 2,000 rpm): Approx. 17.25% • Racing (engine speed is 4,000 rpm): Approx. 19.21%	—
LONGFT1	%	Fuel learning correction amount estimated based on A/F sensor signal	Calculation	• Idle (after warm up): Approx. −6.25% • Racing (engine speed is 2,000 rpm): Approx. −3.9% • Racing (engine speed is 4,000 rpm): Approx. −3.9%	—
LONGFT12	%	Fuel learning correction amount estimated based on HO2S signal	Calculation	• Idle (after warm up): 0%	—
LOW_OIL	Never Detected/Detected	Engine oil pressure condition • Never Detected: Engine oil pressure is high • Detected: Engine oil pressure is low	Calculation	• Displays in the M-MDS but it does not operate.	—
LRN_KCS	—	Knock control system learning value	Calculation	• Displays knock control system learning value	—
M_GEAR*1	Neutral/1st gear/2nd gear/3rd gear/4th gear/5th gear/6th gear/Reverse/Undefined/Auto/In_Progress/YSF/Error	Manual gear position	Calculation	• Displays manual gear position	CAN (2S, 2T)
MAF	g/Sec	Mass air flow input from MAF sensor	Calculation	• Displays MAF	2BE
	V	MAF sensor voltage	Input	• Ignition switched ON (engine off) (MAF: 0.00 g/s {0 lb/min}): Approx. 1.69 V (ECT is 53 °C {127 °F}) • Idle (after warm up) (MAF: 2.50 g/s {0.331 lb/min}): Approx. 1.89 V (ECT is 93 °C {199 °F}) • Racing (engine speed is 2,000 rpm) (MAF: 3.80 g/s {0.503 lb/min}): Approx. 2.02 V (ECT is 95 °C {203 °F})	
MAP	KPa {MPa}, mBar {Bar}, psi, in H20	Manifold absolute pressure input from MAP sensor	Calculation	• Displays MAP	1X
MAP_V	V	MAP sensor voltage	Input	• Ignition switched ON (engine off) (MAP:101 kPa {1.03 kgf/cm^2, 14.6 psi}): Approx. 4.12 V • Idle (after warm up) (MAP: 33 kPa {0.34 kgf/cm^2, 4.8 psi}): Approx. 1.9 V • Racing (engine speed is 2,000 rpm) (MAP: 26 kPa {0.27 kgf/cm^2, 3.8 psi}): Approx. 1.23 V	
MF_CAT1	—	Number of misfires in No.1 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.1 cylinder)	—
MF_CAT_2	—	Number of misfires in No.2 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.2 cylinder)	—
MF_CAT_3	—	Number of misfires in No.3 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.3 cylinder)	—
MF_CAT_4	—	Number of misfires in No.4 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.4 cylinder)	—
MF_CAT_FCC	—	Threshold of malfunction determination for number of misfires (total number in all cylinders) leading to catalytic converter temperature increase	Calculation	• Displays number of misfire determinations (for catalytic converter)	—

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
MF_CAT_TTL	—	Number of misfires (total number in all cylinders) leading to catalytic converter temperature increase	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (total)	—
MF_EMI1	—	Number of misfires in No.1 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.1 cylinder)	—
MF_EMI_2	—	Number of misfires in No.2 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.2 cylinder)	—
MF_EMI_3	—	Number of misfires in No.3 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.3 cylinder)	—
MF_EMI_4	—	Number of misfires in No.4 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.4 cylinder)	—
MF_EMI_FCC	—	Threshold of malfunction determination for number of misfires (total number in all cylinders) under conditions required by emission regulations	Calculation	• Displays number of misfire determinations (for emission)	—
MF_EMI_TTL	—	Number of misfires (total number in all cylinders) under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (total)	—
MIL	Off/On	Check engine light illumination status	Calculation	• Check engine light not illuminated: Off • Check engine light illuminated: On	—
MIL_DIS	Km, ft, mi	Travelled distance since check engine light illuminated	Calculation	• Displays travelled distance since check engine light illuminated	—
MSF_CP	Displays in the M-MDS but it does not operate.				
MSF_ENA	Displays in the M-MDS but it does not operate.				
NEUTRAL_SW1*1	Off/On	Neutral switch No.1 status	Calculation	• Neutral: On • Other than neutral: Off	1BF
NEUTRAL_SW2*1	Off/On	Neutral switch No.2 status	Calculation	• Neutral: On • Other than neutral: Off	1AY
O2_SH_PM_F	Displays in the M-MDS but it does not operate.				
O2_SH_PM_F_CP	Displays in the M-MDS but it does not operate.				
O2_SM_EGSM	Displays in the M-MDS but it does not operate.				
O2_SM_EGSM_CP	Displays in the M-MDS but it does not operate.				
O2S11	μA	A/F sensor current	Input	• Idle (after warm up): 0—50 μA • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): Approx. 3.84 mA	1S
O2S12	V	HO2S voltage	Input	• Idle (after warm up): 0—1.0 V • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): Approx. 0 V	2BK/2BL
OIL_P_DUTY	Displays in the M-MDS but it does not operate.				
OIL_P_SOL	Off/On	Engine oil solenoid valve operation status	Calculation	• ECT below 98 °C {208 °F} and engine speed below 4,000 rpm: On • ECT above 98 °C {208 °F} or engine speed above 4,000 rpm: Off	1CK
OIL_PRES	V	Engine oil pressure	Calculation	• Displays estimated engine oil pressure	—
OIL_TEMP	°C, °F	Estimated engine oil temperature	Calculation	• Displays estimated engine oil temperature	—
PKTC_ENE_LT	Displays in the M-MDS but it does not operate.				
PKTC_ENE_R	Displays in the M-MDS but it does not operate.				
PN_SW*2	Open/Closed	Parking/neutral	Calculation	• Selector lever at P position or N position: Closed • Except above: Open	CAN (2S, 2T)
PRS_TRQ_ACT	%	Actual engine-percent torque	Calculation	• Displays actual engine-percent torque	—
REV_SW*2	Off/On	Back-up light switch status	Calculation	• Selector lever at P position or N position: Closed • Except above: Open	—
RO2FT1	%	Fuel correction amount based on HO2S signal including SHRTFT12 and LONGFT12	Calculation	• Idle (after warm up): Approx. 0% • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): Approx. 3.99%	—
RPM	RPM	Engine speed	Calculation	• Displays engine speed	1F
SEGRP	Displays in the M-MDS but it does not operate.				
SEGRP DSD	Displays in the M-MDS but it does not operate.				
SHRTFT1	%	Fuel feedback correction amount estimated based on A/F sensor signal	Calculation	• Idle (after warm up): Approx. 0% • Racing (engine speed is 2,000 rpm): Approx. 1.56% • Racing (engine speed is 4,000 rpm): Approx. −4.68%	—
SHRTFT12	%	Fuel feedback correction amount estimated based on HO2S signal	Calculation	• Idle (after warm up): 0%	—
SPARKADV	°	Ignition timing	Calculation	• Displays ignition timing	—
Test	Off/On	Test mode	Calculation	• Test mode OFF: Off • Test mode ON: On	—
TH_M	Displays in the M-MDS but it does not operate.				
TH_M_MAX	Displays in the M-MDS but it does not operate.				
TH_M_MIN	Displays in the M-MDS but it does not operate.				
TM_GERT	Displays in the M-MDS but it does not operate.				
TM_GERT_SP	Displays in the M-MDS but it does not operate.				
TM_GEST	Displays in the M-MDS but it does not operate.				
TM_GEST_SP	Displays in the M-MDS but it does not operate.				

Item (definition)	Unit/Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
TP_REL	%	Throttle valve opening angle (relative value) with value at throttle valve fully close timing as the start point	Calculation	• Accelerator pedal released: Approx. 12% • Accelerator pedal fully depressed: Approx. 82%	—
TP_UP_DSD_P	Displays in the M-MDS but it does not operate.				
TP_UP_TEMP	Displays in the M-MDS but it does not operate.				
TP1	%	Throttle valve position No.1	Calculation	• Accelerator pedal released: Approx. 22% • Accelerator pedal fully depressed: Approx. 92%	1O
	V	TP sensor No.1 voltage	Input	• Accelerator pedal released: Approx. 1.11 V • Accelerator pedal fully depressed: Approx. 4.59 V	
TP2	%	Throttle valve position No.2	Calculation	• Accelerator pedal released: Approx. 22% • Accelerator pedal fully depressed: Approx. 92%	1V
	V	TP sensor No.2 voltage	Input	• Accelerator pedal released: Approx. 3.92 V • Accelerator pedal fully depressed: Approx. 0.41 V	
TPCT	V	TP sensor No.1 minimum voltage at CTP	Calculation	• Ignition switched ON (engine off): Approx. 0.5 V	1O
TPCT2	V	TP sensor No.2 maximum voltage at CTP	Calculation	• Ignition switched ON (engine off): Approx. 4.5 V	1V
TT_CPS_AT_LT	Displays in the M-MDS but it does not operate.				
TT_CPS_AT_R	Displays in the M-MDS but it does not operate.				
TT_DTRV_LT	Displays in the M-MDS but it does not operate.				
TT_DTRV_R	Displays in the M-MDS but it does not operate.				
TT_ERT_LT	Displays in the M-MDS but it does not operate.				
TT_ERT_R	Displays in the M-MDS but it does not operate.				
TT_FCSM_R	Displays in the M-MDS but it does not operate.				
TT_IE_RT_LT	Displays in the M-MDS but it does not operate.				
TT_IE_RT_R	Displays in the M-MDS but it does not operate.				
TT_IPS_AT_LT	Displays in the M-MDS but it does not operate.				
TT_IPS_AT_R	Displays in the M-MDS but it does not operate.				
TT_PS_AT_LT	Displays in the M-MDS but it does not operate.				
TT_PS_AT_R	Displays in the M-MDS but it does not operate.				
VEH_FEL_RATE	Displays in the M-MDS but it does not operate.				
VEH_ODO	Displays in the M-MDS but it does not operate.				
VEV_OIL_P	Displays in the M-MDS but it does not operate.				
VPWR	V	Battery positive voltage	Input	• Displays battery voltage	2A
VSS	KPH, MPH	Vehicle speed	Calculation	• Displays vehicle speed	CAN (2S, 2T)
VT_IN_ACT	°	Actual intake valve timing	Calculation	• Displays actual intake valve timing **Idle (after warm up)** — Approx. 57 ° **Racing (engine speed is 2,000 rpm)** — Approx. 84 °	1BT
VT_IN_DES	°	Desired intake valve timing	Calculation	• Displays desired intake valve timing **Idle (after warm up)** — Approx. 0 ° **Racing (engine speed is 2,000 rpm)** — Approx. 29 °	—
VT_EX_ACT	°	Actual exhaust valve timing	Calculation	• Displays actual exhaust valve timing	1BZ
VT_EX_DES	°	Desired exhaust valve timing	Calculation	• Displays desired exhaust valve timing	—
VT_EX_DUTY	%	OCV control duty value	Calculation	• Idle (after warm up): 0% • Racing (engine speed is 2,000 rpm): Approx. 40%	1CO
WGP	Displays in the M-MDS but it does not operate.				
WGV_ACT	Displays in the M-MDS but it does not operate.				
WGV_DSD	Displays in the M-MDS but it does not operate.				

*1 :MTX
*2 :ATX
*3 :With i-ELOOP
*4 :Without i-ELOOP
*5 :SKYACTIV-G 1.5